Integrated Molecular Signature of BRCA Subtypes

Data Integration Analysis (DIABLO) of mRNA and miRNA

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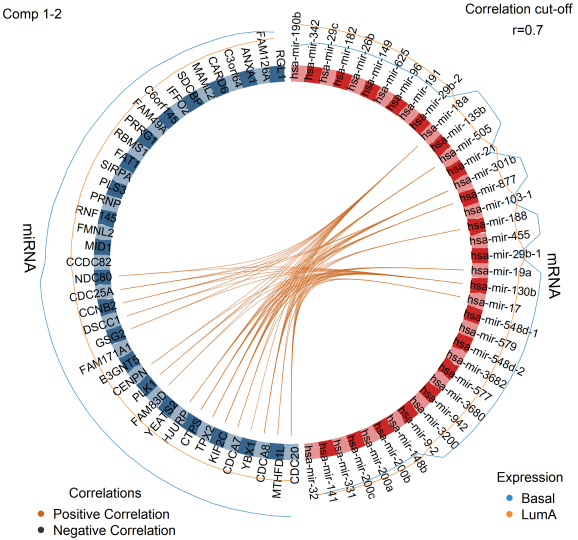
Methodology: DIABLO Framework

- **Data Source**: Matched TCGA-BRCA mRNA and miRNA datasets.
- **Objective**: Identify a multi-modal signature for Basal and Luminal A.
- **Feature Selection**: Top 20 features per omics layer selected for discrimination.

Global Regulatory Overview (F

Multi-Omics Connectivity

- **Circos Plot**: Visualizes regulatory crosstalk between miRNAs and genes.
- **Confidence**: Focuses on strong correlations where $r > 0.7$.
- **Alignment**: Reveals unique expression patterns for each subtype.

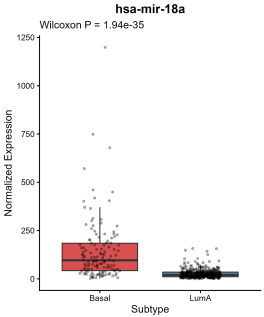


Integrated Molecular Signature (Figure 4)

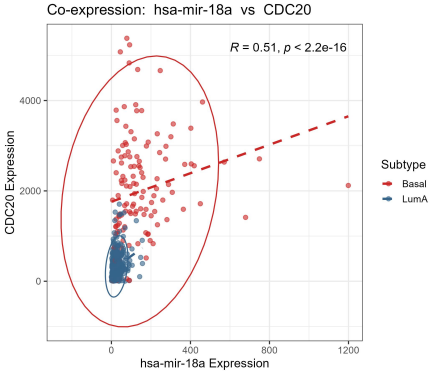
Expression Pattern Analysis

- **Z-Score Scaling**: Visualizes relative expression across the cohort.
- **Basal Signature**: Distinct red cluster indicates significant upregulation.
- **Luminal A Signature**: Characteristic blue cluster showing relative downregulation.

Top Regulatory Drivers (Figures 2 & 3)



***hsa-mir-18a* Expression**



miRNA-Gene Correlation

Conclusion

- Identified a robust 40-feature multi-omics signature.
- **Key Drivers**: *hsa-mir-18a* and cell-cycle genes like *CDC20*.
- Multi-omics provides superior classification power over single-layer analysis.